

PRACTICAL – 8

CODE:-

```
#include<stdio.h>

int main()
{
    int ref[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3] = {-1,-1,-1};
    int time[3] = {0,0,0};
    int n = 12;
    int i,j,pos,least;
    int hit = 0, fault = 0;
    int counter = 0;
    int found;
    printf("\n===== LRU PAGE REPLACEMENT =====\n");
    printf("\nReference String : ");
    for(i=0;i<n;i++)
        printf("%d ", ref[i]);
    printf("\n\nFrame Status:\n");
    for(i=0;i<n;i++)
    {
        found = 0;
        for(j=0;j<3;j++)
        {
            if(frames[j] == ref[i])
            {
                hit++;
                counter++;
            }
        }
    }
}
```

```

        time[j] = counter;

        found = 1;

        printf("\nPage %d -> HIT | ", ref[i]);

        break;
    }
}

```

```

if(found == 0)
{
    least = time[0];

    pos = 0;

    for(j=1;j<3;j++)
    {
        if(time[j] < least)
        {
            least = time[j];

            pos = j;
        }
    }

    frames[pos] = ref[i];

    counter++;

    time[pos] = counter;

    fault++;

    printf("\nPage %d -> FAULT | ", ref[i]);
}

for(j=0;j<3;j++)
    printf("%d ", frames[j]);
}

```

```
printf("\n\nTotal References = %d", n);
printf("\nTotal Page Hits = %d", hit);
printf("\nTotal Page Faults = %d", fault);

float hitp = (float)hit/n * 100;
float faultp = (float)fault/n * 100;

printf("\nHit Percentage = %.2f%%", hitp);
printf("\nFault Percentage = %.2f%%", faultp);

return 0;
}
```

OUTPUT -

Output

===== LRU PAGE REPLACEMENT =====

Reference String : 1 2 3 4 1 2 5 1 2 3 4 5

Frame Status:

Page 1 -> FAULT | 1 -1 -1

Page 2 -> FAULT | 1 2 -1

Page 3 -> FAULT | 1 2 3

Page 4 -> FAULT | 4 2 3

Page 1 -> FAULT | 4 1 3

Page 2 -> FAULT | 4 1 2

Page 5 -> FAULT | 5 1 2

Page 1 -> HIT | 5 1 2

Page 2 -> HIT | 5 1 2

Page 3 -> FAULT | 3 1 2

Page 4 -> FAULT | 3 4 2

Page 5 -> FAULT | 3 4 5

Total References = 12

Total Page Hits = 2

Total Page Faults = 10

Hit Percentage = 16.67%

Fault Percentage = 83.33%

=== Code Execution Successful ===